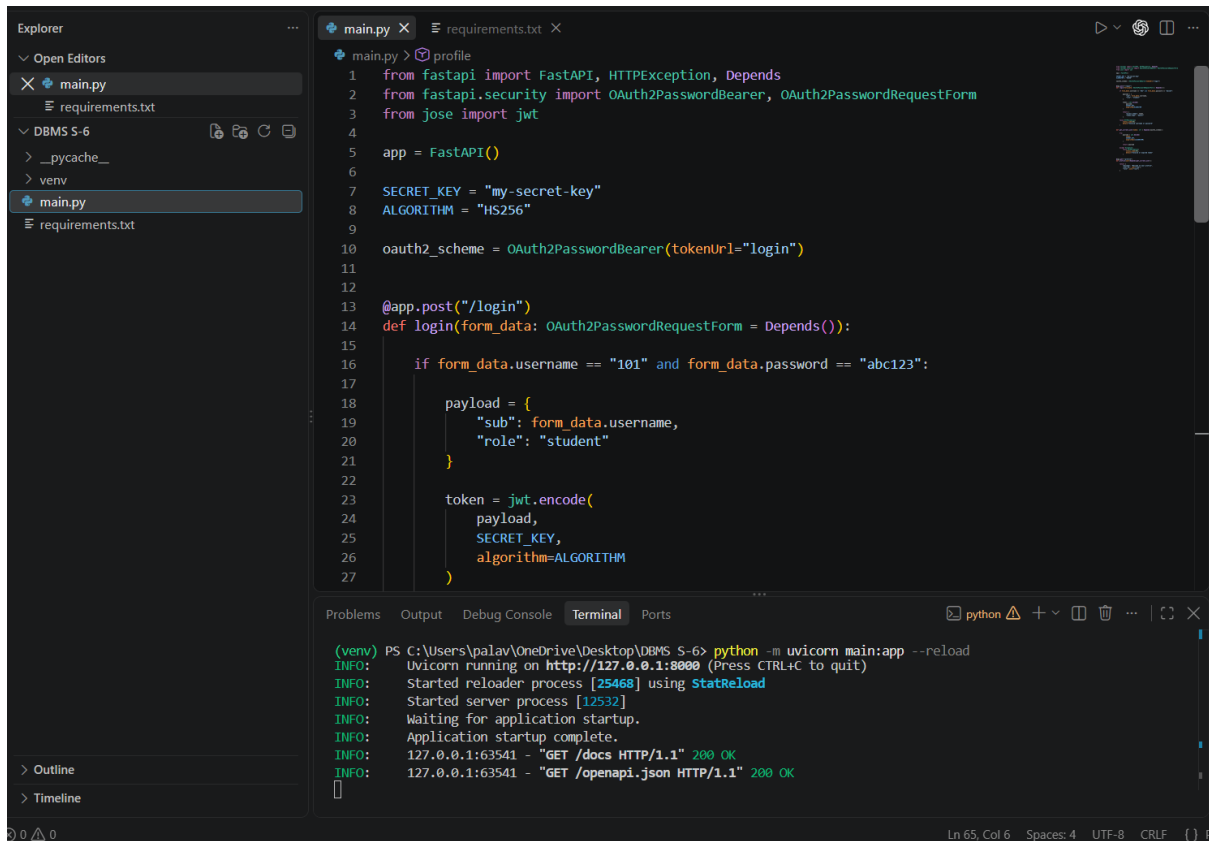


2520090016

P.Rahul

SKILL-6



The screenshot shows a VS Code editor with a project named 'DBMS S-6'. The Explorer sidebar on the left shows the file structure with 'main.py' and 'requirements.txt' selected. The main editor displays the code in 'main.py', which imports FastAPI, OAuth2PasswordBearer, OAuth2PasswordRequestForm, and jwt. It defines a FastAPI app, sets a secret key and algorithm, and implements a login endpoint. The login endpoint checks for a specific username and password, and if successful, it generates a JWT token. The terminal at the bottom shows the command to run the application using uvicorn, and the output indicates that the server is running on http://127.0.0.1:8000 and has successfully processed GET requests to /docs and /openapi.json.

```
1 from fastapi import FastAPI, HTTPException, Depends
2 from fastapi.security import OAuth2PasswordBearer, OAuth2PasswordRequestForm
3 from jose import jwt
4
5 app = FastAPI()
6
7 SECRET_KEY = "my-secret-key"
8 ALGORITHM = "HS256"
9
10 oauth2_scheme = OAuth2PasswordBearer(tokenUrl="login")
11
12
13 @app.post("/login")
14 def login(form_data: OAuth2PasswordRequestForm = Depends()):
15     if form_data.username == "101" and form_data.password == "abc123":
16         payload = {
17             "sub": form_data.username,
18             "role": "student"
19         }
20         token = jwt.encode(
21             payload,
22             SECRET_KEY,
23             algorithm=ALGORITHM
24         )
25     return {"token": token}
```

Problems Output Debug Console Terminal Ports

(venv) PS C:\Users\palav\OneDrive\Desktop\DBMS S-6> python -m uvicorn main:app --reload

INFO: Uvicorn running on http://127.0.0.1:8000 (Press CTRL+C to quit)

INFO: Started reloader process [25468] using StatReload

INFO: Started server process [12532]

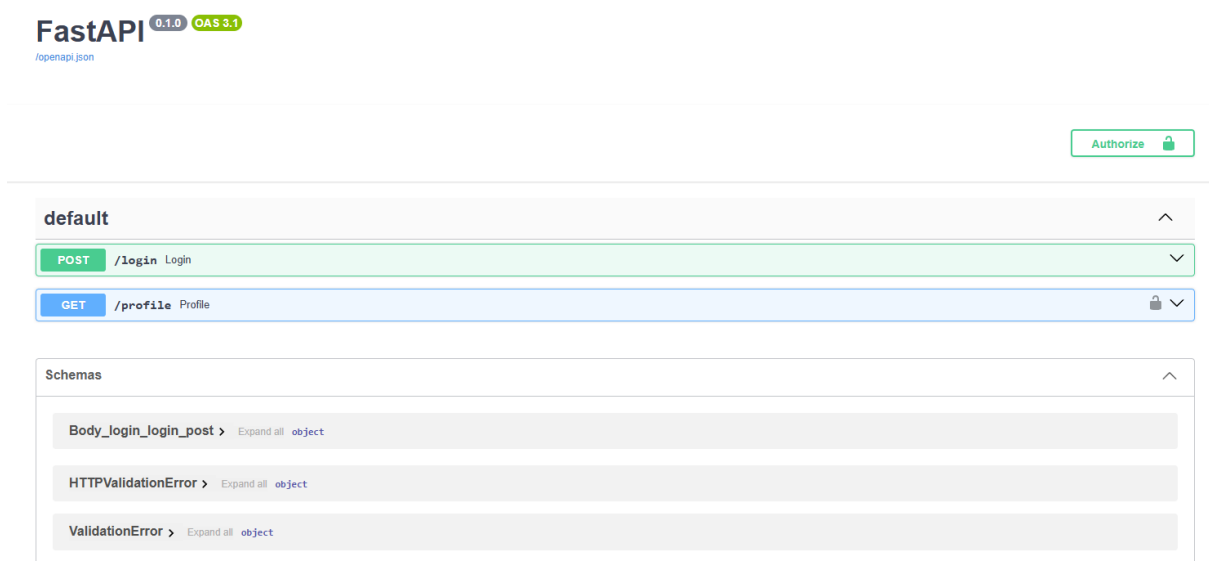
INFO: Waiting for application startup.

INFO: Application startup complete.

INFO: 127.0.0.1:63541 - "GET /docs HTTP/1.1" 200 OK

INFO: 127.0.0.1:63541 - "GET /openapi.json HTTP/1.1" 200 OK

Swagger:



POST/login:

default

POST /login Login

Parameters

No parameters

Request body required

application/x-www-form-urlencoded

grant_type

string | (string | null)

pattern: ^password\$

password

☐ Send empty value

username * required

string

101

password * required

string(\$password)

scope

string

scope

☒ Send empty value

client_id

string | (string | null)

string

☐ Send empty value

client_secret

string | (string | null)(\$password)

☐ Send empty value

Execute

Clear

Curl

```
curl -X 'POST' \
'http://127.0.0.1:8000/login' \
-H 'accept: */*' \
-H 'Content-type: application/x-www-form-urlencoded' \
-d 'grant_type=password&username=101&password=abc123&scope=&client_id=string&client_secret=*****'
```

Request URL

```
http://127.0.0.1:8000/login
```

Server response

Code

Details

200

Response body

```
{
  "access_token": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJzdWIiOiIxMDEiLCJyb2kiOiJ1b3R1ZGVudCJ9.NHq4dLP6iQ5BFQ1iKmdohvLOjbLURdW4aRQ6W1zI",
  "token_type": "bearer"
}
```

Response headers

```
content-length: 162
content-type: application/json
date: Fri, 18 Sep 2026 16:31:15 GMT
server: uvicorn
```

Responses

Code

Description

Links

200

Successful Response

No links

Media type

application/json

Controls

Accept header

Example Value

Schema

"string"

422

Validation Error

No links

Media type

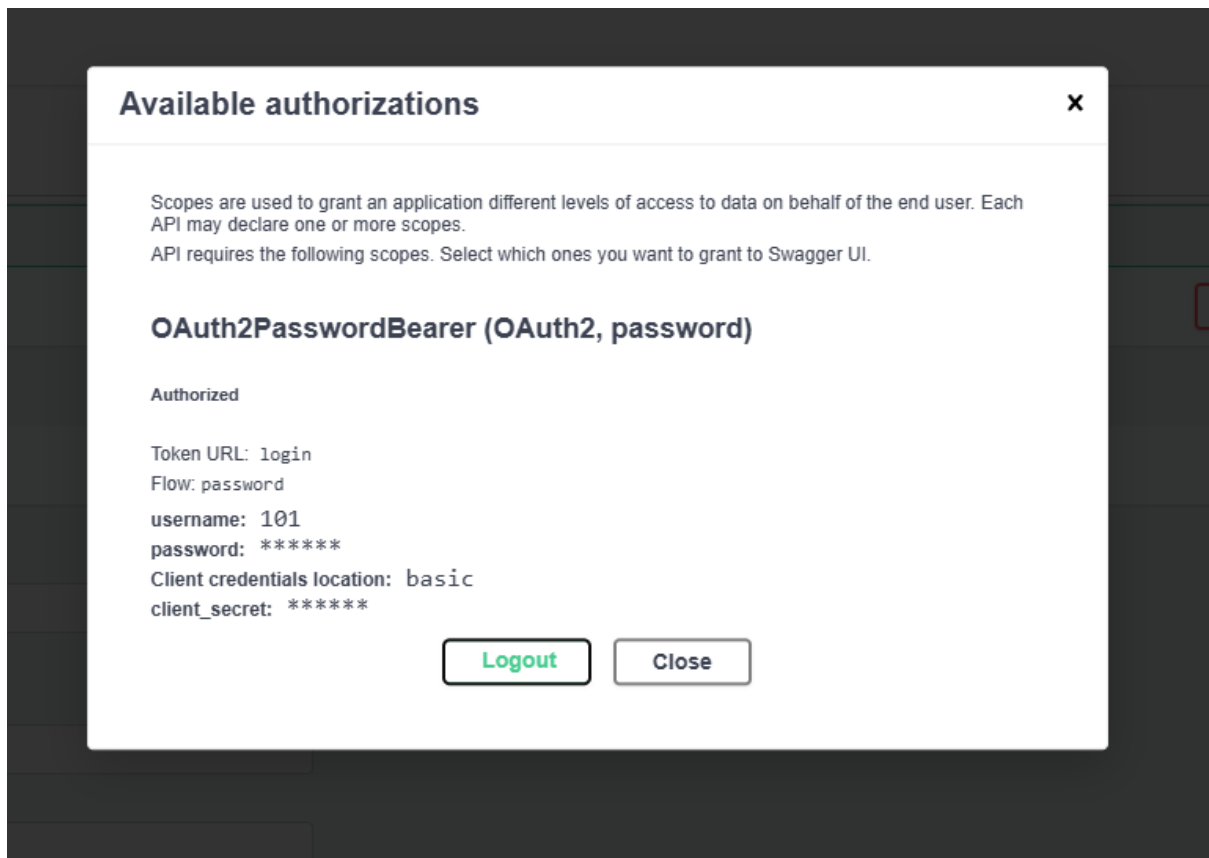
application/json

Example Value

Schema

{
 "detail": [

Authorization:



Test /profile:

